Problem 7-6

a) Attached in matlab file
b) If I were to take more points, I would take them in between 0-15 min because that is where the concentration changes the fastest, which is indicated by the largest slope.
c) If I were to repeat the experiment, I would take values at multiple temperatures to estimate activation energy and see the temperature dependency of the reaction, along with taking samples more frequently to have more points and limit some uncertainties.
d) I think there was a dilution error because that point falls below the regression line and if you remove the point, R^2 increases, and k and alpha values decrease,

Problem 7-7

a) The total pressure at t=0 is equal to the initial partial pressure of dimethyl ether, since no decomposition has occurred yet. It can be estimated from the stoichiometry. For every 1 mole dimethyl ether decomposed, 3 moles of product are generated. So total moles equals $n=n_0(1-x)+3n_0x = n_0(1+2x)$ and since volume and T are constant, $P_t=P_0(1+2x)$, when the reaction is completed x=1, so P(infinite) = $3P_0$. So, P_0 can be estimated at P at infinity/3 = 310.3 mmHg.
b) The best fit for reaction order is first order, and k is found to be 0.00048 min^{-1}.
c) If I were to obtain more data, I would take more data at early times, maintain constant temperature and volume and run at different initial pressures to confirm order isn't pressure dependent.
d) If the reaction is run at a higher temperature, k would increase, and the curve would reach equilibrium sooner and if the reaction is run at a lower temperature, k would decrease and it would take more time to reach equilibrium.